

OPERATING SYSTEMS

WEEK-3 Practical

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Section no:- 9

1. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

Program:-

```
GNU nano 8.7.1
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<sys/wait.h>
int main() {
    pid_t pid;
    pid = fork();
    if(pid > 0){
        printf("Parent Process\n");
        printf("%d\n",getpid());
        printf("%d\n",getppid());
    }
    else if(pid == 0){
        printf("Child Process\n");
        printf("%d\n",getpid());
        printf("%d\n",getppid());
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else{
        wait(NULL);
        printf("Parent resumes after child finishes.\n");
    }
    return 0;
}
```

Output:-

```
Parent Process
793
329
Child Process
794
328
dhanu@dhnpc:~$ total 144
-rw-r--r-- 1 dhanu dhanu 506 Jul 28 05:32 Arithmetic.c
-rw-r--r-- 1 dhanu dhanu 156 Aug 4 14:14 Excel.c
-rw-r--r-- 1 dhanu dhanu 0 Aug 3 08:20 Excel2.c
-rw-r--r-- 1 dhanu dhanu 0 Aug 4 14:12 Excel.c
-rw-r--r-- 1 dhanu dhanu 242 Aug 4 14:17 Exec2.c
-rw-r--r-- 1 dhanu dhanu 352 Aug 17 06:52 Exec3.c
-rw-r--r-- 1 dhanu dhanu 426 Aug 17 07:03 Exec4.c
-rw-r--r-- 1 dhanu dhanu 0 Aug 3 06:56 Exec1.c
-rw-r--r-- 1 dhanu dhanu 172 Jul 30 08:03 Process1.c
-rw-r--r-- 1 dhanu dhanu 215 Aug 3 02:28 Process2.c
-rw-r--r-- 1 dhanu dhanu 154 Aug 1 09:19 Process2.c.save
-rw-r--r-- 1 dhanu dhanu 184 Aug 3 02:30 Process3.c
-rw-r--r-- 1 dhanu dhanu 550 Aug 3 02:38 Process4.c
-rw-r--r-- 1 dhanu dhanu 550 Aug 1 10:17 Process4.c.save
-rw-r--r-- 1 dhanu dhanu 185 Aug 10 15:49 Sys1.c
-rw-r--r-- 1 dhanu dhanu 324 Aug 10 15:54 Sys2.c
-rw-r--r-- 1 dhanu dhanu 295 Aug 11 05:55 Sys3.c
-rw-r--r-- 1 dhanu dhanu 1 Aug 11 05:53 Z01.c
-rwxr-xr-x 1 dhanu dhanu 16208 Aug 17 07:03 a.out
-rwxr-xr-x 1 dhanu dhanu 16056 Jul 28 04:42 add
-rw-r--r-- 1 dhanu dhanu 145 Jul 28 04:41 addition.c
-rwxr-xr-x 1 dhanu dhanu 16064 Jul 28 05:33 ari
-rwxr-xr-x 1 dhanu dhanu 16008 Aug 3 08:00 exe
-rwxr-xr-x 1 dhanu dhanu 16248 Aug 3 02:38 pro
dhanu@dhnpc:~$ |
```